AI Ethics Decision Simulator Using C

1. Introduction

The AI Ethics Decision Simulator is a console-based application developed using the C programming language. This project simulates real-world ethical dilemmas and analyzes user decisions based on fundamental ethical principles used in Artificial Intelligence systems.

The project aims to bridge theoretical knowledge and practical implementation by allowing users to interact with different scenarios and observe how ethical reasoning can be applied through simple rule-based logic.

2. Objectives

The main objectives of this project are:

- To understand basic AI ethical decision-making
- To apply programming logic in real-world scenarios
- To develop structured and modular C programs
- To improve problem-solving and analytical thinking skills

3. Features

The key features of the AI Ethics Decision Simulator are:

- Interactive menu-driven system
- Multiple real-world ethical scenarios
- User decision-based analysis
- Rule-based ethical evaluation (if-else logic)
- Display of ethical theory with explanation
- Simple scoring system (Good / Neutral / Risky)
- Continuous execution using loop (repeat option)
- Beginner-friendly and easy to understandThe system is designed using a structured flow-based approach.

4. Tools & Technologies

- Programming Language: C
- Compiler: GCC / CodeBlocks
- Platform: Console-based application

5. System Design

- The system is designed using a structured flow-based approach.
- Design Overview:
- The program starts with a main menu
- User selects a scenario
- The system displays the problem and options
- User makes a decision
- The system applies predefined rules
- Ethical analysis is generated
- User can return to menu or exit

Flow of the system:

- Start program
- Display main menu
- User selects scenario
- Show scenario description
- Take user decision
- Apply rule-based logic (if-else)
- Generate ethical analysis
- Display result and score
- Ask to return to menu or exit
- End program

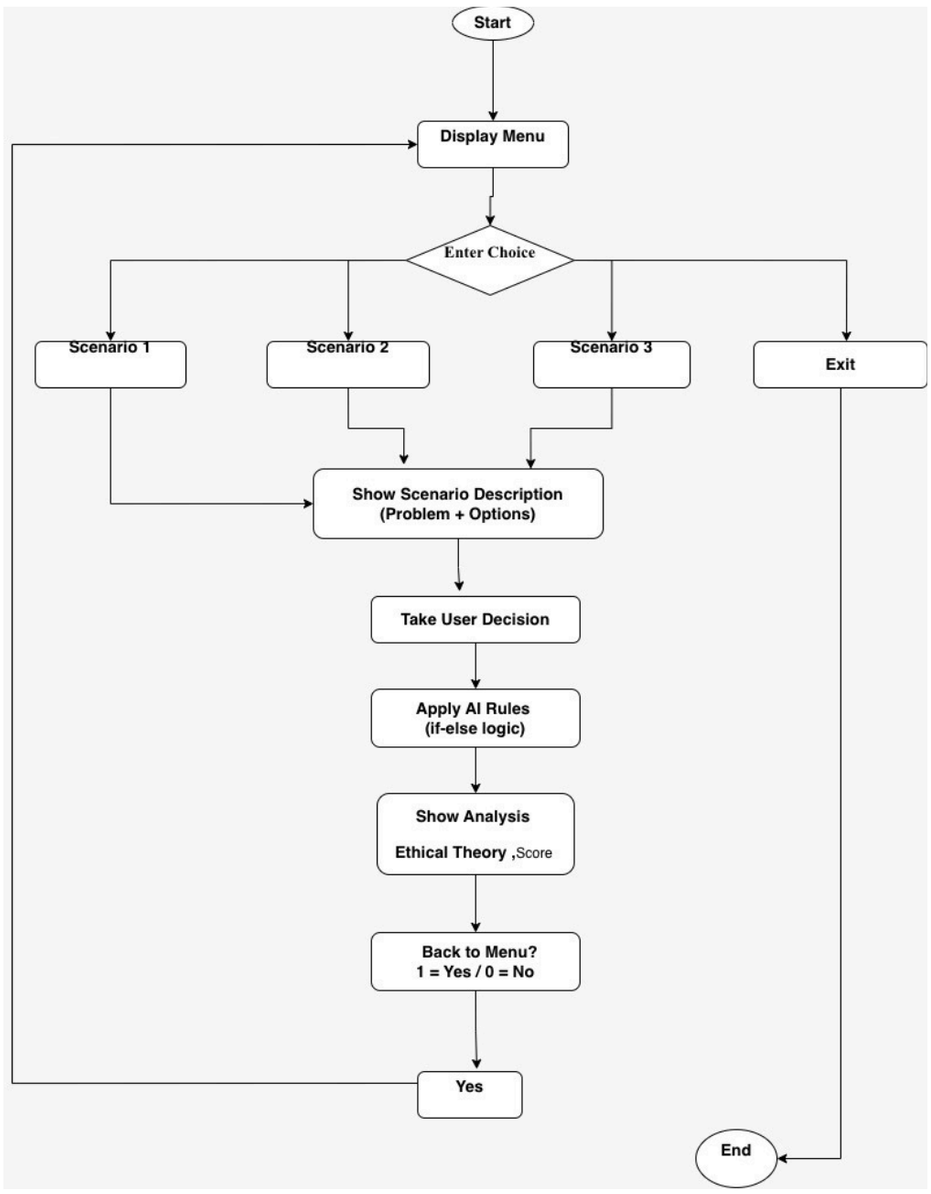


Figure 1: Flowchart of AI Ethics Decision Simulator

6. Implementation

Scenario 1: Self-Driving Car

A critical situation where a car must decide between saving passengers or pedestrians.

Focus: Individual Rights vs Utilitarian Ethics

Scenario 2: Medical Decision

A scenario involving limited medical resources (one ventilator).

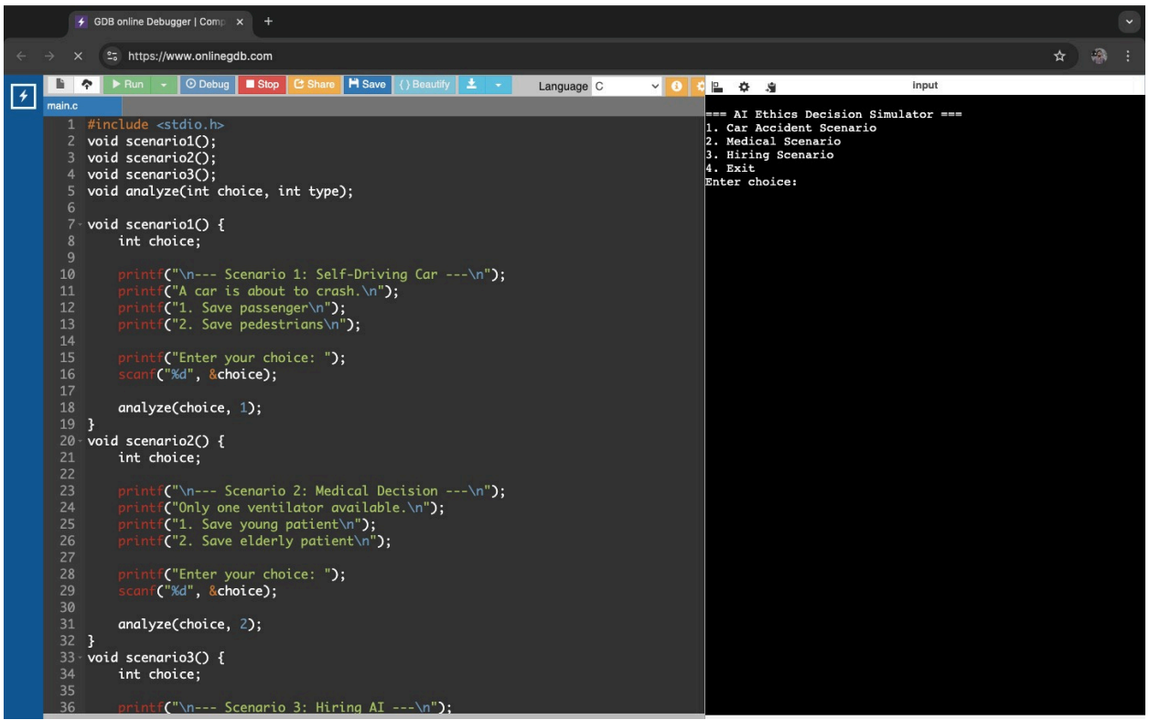
Focus: Fairness vs Long-term benefit

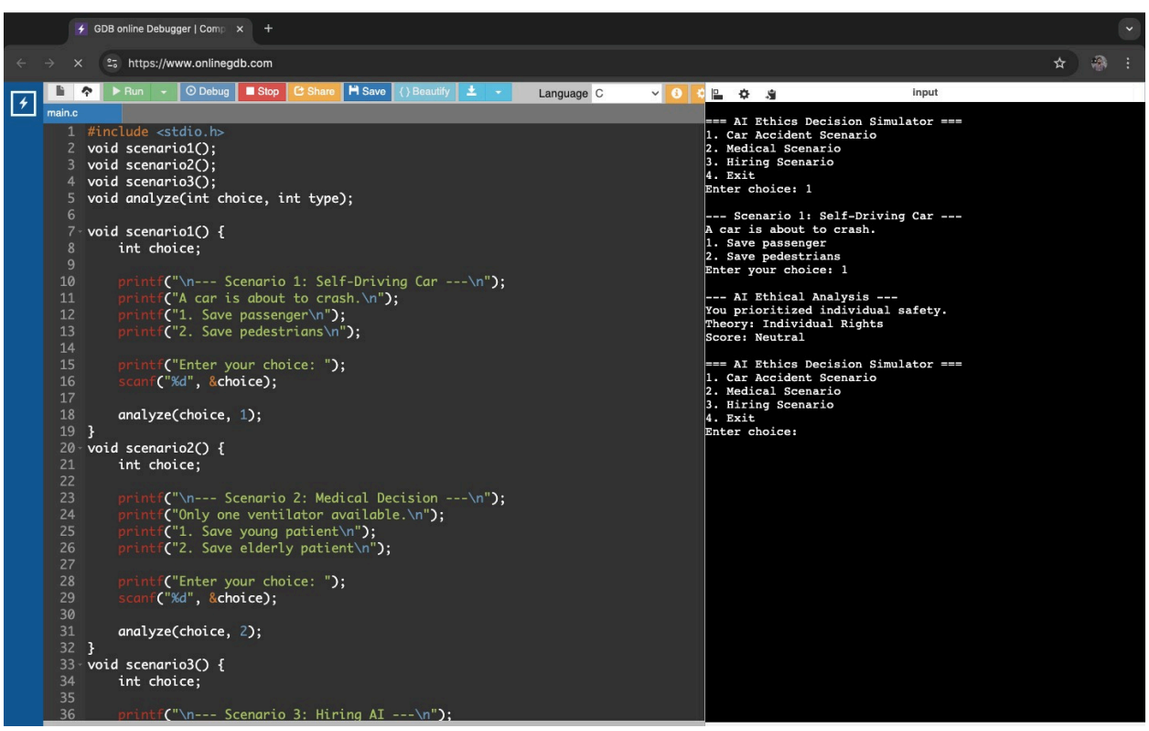
Scenario 3: Hiring AI

A recruitment decision involving bias and fairness.

Focus: Ethical AI and Bias Risk

7. Sample Input/Output





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main.c

```c
#include <stdio.h>
void scenario1();
void scenario2();
void scenario3();
void analyze(int choice, int type);

void scenario1() {
    int choice;

    printf("\n--- Scenario 1: Self-Driving Car ---\n");
    printf("A car is about to crash.\n");
    printf("1. Save passenger\n");
    printf("2. Save pedestrians\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    analyze(choice, 1);
}
void scenario2() {
    int choice;

    printf("\n--- Scenario 2: Medical Decision ---\n");
    printf("Only one ventilator available.\n");
    printf("1. Save young patient\n");
    printf("2. Save elderly patient\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    analyze(choice, 2);
}
void scenario3() {
    int choice;

    printf("\n--- Scenario 3: Hiring AI ---\n");
```

Input

```
=== AI Ethics Decision Simulator ===
1. Car Accident Scenario
2. Medical Scenario
3. Hiring Scenario
4. Exit
Enter choice: 1

--- Scenario 1: Self-Driving Car ---
A car is about to crash.
1. Save passenger
2. Save pedestrians
Enter your choice: 1

--- AI Ethical Analysis ---
You prioritized individual safety.
Theory: Individual Rights
Score: Neutral

=== AI Ethics Decision Simulator ===
1. Car Accident Scenario
2. Medical Scenario
3. Hiring Scenario
4. Exit
Enter choice: 2

--- Scenario 2: Medical Decision ---
Only one ventilator available.
1. Save young patient
2. Save elderly patient
Enter your choice: 1

--- AI Ethical Analysis ---
You chose long-term potential.
Theory: Utilitarian Ethics
Score: Good

=== AI Ethics Decision Simulator ===
1. Car Accident Scenario
2. Medical Scenario
3. Hiring Scenario
4. Exit
Enter choice:
```

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    scanf("%d", &choice);

    analyze(choice, 1);
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    printf("\n--- Scenario 2: Medical Decision ---\n");
    printf("Only one ventilator available.\n");
    printf("1. Save young patient\n");
    printf("2. Save elderly patient\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    analyze(choice, 2);
}
void scenario3() {
    int choice;

    printf("\n--- Scenario 3: Hiring AI ---\n");
```

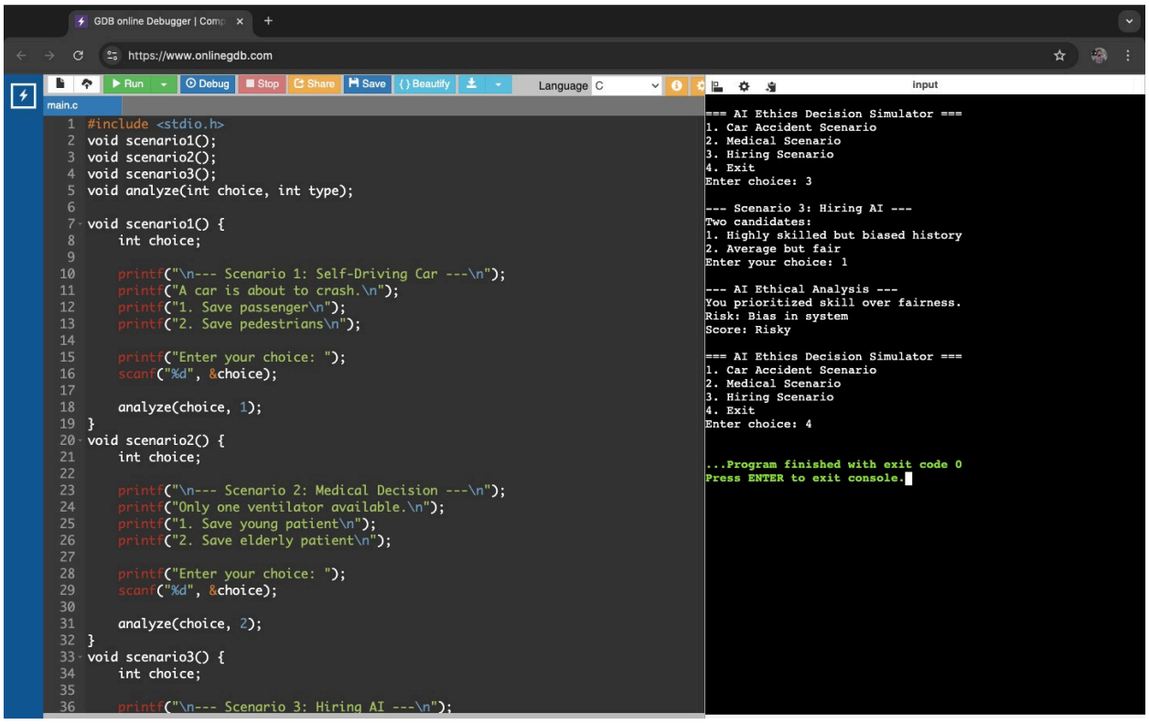
Input

```
=== AI Ethics Decision Simulator ===
1. Car Accident Scenario
2. Medical Scenario
3. Hiring Scenario
4. Exit
Enter choice: 3

--- Scenario 3: Hiring AI ---
Two candidates:
1. Highly skilled but biased history
2. Average but fair
Enter your choice: 1

--- AI Ethical Analysis ---
You prioritized skill over fairness.
Risk: Bias in system
Score: Risky

=== AI Ethics Decision Simulator ===
1. Car Accident Scenario
2. Medical Scenario
3. Hiring Scenario
4. Exit
Enter choice:
```



8. Ethical Theories Applied

The system evaluates decisions based on:

- Utilitarian Ethics (maximizing overall benefit)
- Individual Rights (protecting individuals)
- Fairness (equal treatment)
- Bias Awareness in AI systems

9. Conclusion

The AI Ethics Decision Simulator project demonstrates how ethical decision-making concepts can be implemented using basic programming techniques. It reflects the integration of problem-solving skills, software design principles, and ethical awareness.

This project serves as a strong foundation for future development in Artificial Intelligence and Software Engineering.

10. Future Improvements

- Add more complex and diverse scenarios
- Implement a scoring system with history tracking
- Develop a graphical user interface (GUI)
- Introduce basic machine learning concepts
- Store and analyze user decisions

11. Author

Name: Md Tasnimul Islam Shad

Project Title: AI Ethics Decision Simulator
Programming Language: C

12. Learning Outcomes / Course Integration

Through completing various professional courses, the following skills were developed and applied in this project:

From Microsoft: Problem-solving, algorithmic thinking, and structured programming.

From Google: Version control, project management, and workflow organization.

From IBM: Software development concepts, modular design, and system understanding.

From Stanford University: Understanding of AI ethics and its real-world implications.

This project shows how basic knowledge from different subjects can be combined to create a simple and useful application.